Calendar-Time Maternal Genealogies: Shared Ancestry and Cohort Dispersion

Joint depth–collision limits and explicit whole-cohort results

Manuscript for scholarly review

Abstract

How many maternal generations separate two people from the same date in the past? Different ages at childbirth produce different ancestral depths, but the two paths may also encounter the same woman and hence cease to be independent. We study this distinction in an explicitly specified stochastic pedigree, in which maternal age and ancestral identity are both represented. For a two-age model, we establish a joint functional limit for the depth difference and first shared-ancestor time: a Gaussian fluctuation stopped at an independently arising collision time. Finite demographic regimes yield a corresponding conditional Gaussian–Cox description. We then derive simultaneous depth bounds for an entire cohort and an explicit asymptotic formula for its *expected* mean pairwise depth gap. In one illustrative parameter setting, accounting for common ancestors reduces the theoretical coefficient by approximately 13% relative to independent renewal paths. Exact ancestral-partition identities describe how shared descendants affect cohort statistics. These are model-based mathematical results, not estimates from historical human pedigrees. The broad spatially heterogeneous dense–sparse programme, and the originality of the combined framework beyond the existing specialist literature, remain separate questions.

Keywords: maternal genealogy; calendar-time depth; common ancestors; coalescence; renewal processes; Gaussian–Cox limit; cohort dispersion.

For readers outside probability theory

Imagine tracing two maternal family lines backwards. Even when two people are born on the same day, the number of mother-to-child links needed to reach an earlier century can differ, because one family's generations may be shorter than another's. There is a further complication: if both lines reach the same maternal ancestor, their subsequent histories are shared, not independent.

This paper separates those effects in a deliberately simplified mathematical population. It proves how common ancestry changes a *theoretical expected difference* across an entire cohort; it does not calculate a historical difference between real people. The figures explain the mechanism and its size in one model. The equations make the assumptions testable. The companion 1,406-page technical volume shows the larger mathematical programme and its proofs, with precise navigation at the end of this paper.

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1 Question, context, and scope

A maternal genealogical path is a sequence of individuals, with each backward edge recording the age at which a mother gave birth. Counting how many such edges are required to reach a fixed calendar cut is a renewal problem. Studying two contemporaneous descendants changes the problem: the two paths can select the same historical woman. From the first coalescence onward they have a shared ancestry, so their future calendar-time depth increments cannot be independent.

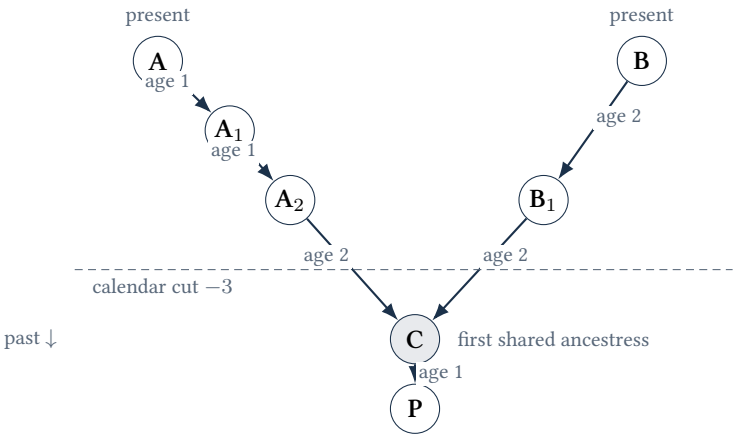


Figure 1: A small *illustrative* pedigree in arbitrary calendar units. The two descendants reach the cut -3 after different numbers of maternal links (three for A, two for B), then share the same woman C at time -4. From C backwards their ancestral identities coincide. This diagram is not a historical pedigree or a simulation result.

Our central question is: which conclusions about pairwise or whole-cohort depth dispersion can be derived from the marginal renewal laws, and which require a model of ancestral *identities*? We separate three levels. First, an explicitly specified, two-age stochastic pedigree admits a full probabilistic proof. Second, a finite ancestral partition gives exact formulas for dispersion and extremes conditional on a marking law. Third, a general historical model can use these insights only after satisfying stronger uniform and non-anticipative estimates. We do not cross the third boundary by fiat. Figure 1 shows the identity mechanism; Figure 2 shows its quantitative consequence in one illustrative model.

Prior literature. Functional limits for renewal counts are classical; so are Kingman-type limits for age-structured and spatially structured populations [6, 7, 8]. Large-family genealogies and the need for hybrid exact/approximate treatments have also been investigated [11]. None of these mechanisms is claimed to have been invented here. Our restricted theorem makes one joint calendar-depth/stopping assertion fully explicit and proves it within the particular maternal-identity construction below. A universal originality claim would require a more exhaustive specialist review than the comparison presently available.

The companion corpus M0–M6c comprises general statements involving Markovian maternal ages, historical interfaces, spatial classes, reproductive weights, ancestral partitions, dense–sparse handoff, and scenario-qualified compilation. Internal source identifiers are given in Section 13. This article is *not* a replacement for the general proofs, whose remaining conditions are described precisely in Section 12.

1.1 Contributions and their precise evidentiary status

The results of this manuscript have different mathematical roles. The two-lineage limit (Theorems 4.1 and 5.1) concerns an explicit stochastic pedigree; it is *not* a claim about observed humans. The occupation estimate and Proposition 4.4 quantify an asymptotic calculation within that model. The ancestral partition identities (Propositions 8.1 and 8.2) are exact finite-population statements. The new full-cohort

results (Theorems 9.2, 9.4 and 9.6) are proved for the explicit single-pool model; the first theorem controls all observed leaves simultaneously, the second provides only an *expected* pair statistic, and the third controls only simultaneous merger multiplicity in growing samples. The structural-mark transfer (Theorem 10.1) is likewise exact, but becomes an asymptotic genealogical assertion only if its uniform approximation premise is proved. Finally, Section 11 describes the methodological contract specified by M6a–M6c, not an empirical validation.

Statement	Scientific content	Logical status
Two-lineage limit	Joint Gaussian fluctuations and identity-collision stopping	Proved in model
Finite regimes	Conditional Gaussian–Poisson; Cox after mixing	Proved in model
Dependent marks	Exact effect of block-mark covariance on pair disparity	Finite identity
Full cohort	Uniform range bound and expected mean gap in a one-pool pedigree	Proved in model
Growing sample	Probability bound for simultaneous multiple collisions	Proved in model
Cohort transfer	Quantiles, disparity and range from uniform marked-leaf error	Conditional theorem
Historical compilation	Data requirements and admissible scenario provenance	Methodological contract

1.2 Relation to classical results

Markov-renewal processes and counting-process invariance principles are classical [1, 6]. The coalescent was introduced in foundational form by Kingman [2]; age-ordered genealogical convergence already appears in Donnelly and Joyce [3], and age-structured ancestral approximations appear in Sagitov and Jagers [7]. Time-varying reproduction and population sizes are studied by Möhle [5], whereas random-environment and spatial structure are investigated by Sagitov, Jagers and Vatutin [8]. Hybrid exact/asymptotic approximations for large samples have substantial prior art [11]. Pedigree dependence and rare large-family mergers require separate caution, as discussed by Wakeley and collaborators [9, 10]. Joint covariance structures for tree statistics have been studied systematically [4]. Accordingly, neither the Gaussian limit, Poisson collision mechanism, Cox mixing, nor shared ancestry *separately* is claimed to originate here. The mathematical value of the present formulation is its transparent coupling of calendar-depth counts and identified maternal vertices, a checkable stopping operation, and the separation of finite cohort identities from the additional uniform probabilistic estimates they would require. Whether that combination warrants journal publication is a separate, open originality question.

2 An exact maternal population model

Fix $p \in (0,1)$ and $\nu > 0$. For each integer n let $N_n = \lfloor \nu n \rfloor \geq 2$. There are exactly N_n distinguishable female individuals at every integer birth date $-t$, $t \geq 0$. For every individual v born at $-t$, independently across distinct v, specify a maternal age

$$\mathbb{P}(A(v) = 1) = p, \qquad \mathbb{P}(A(v) = 2) = 1 - p, \tag{1}$$

and select a mother uniformly among the N_n individuals born at $-(t + A(v))$. This defines a random, locally finite, acyclic maternal graph: dates strictly decrease along mother edges. An individual may have several daughters, and daughters born at different dates can select the same mother. The age variable attached to a *vertex* is used only when its outgoing maternal edge is explored.

Choose two distinct individuals from the birth-date-zero cohort. Their real maternal lineages agree after the first shared individual. To facilitate analysis, construct two *ghost* lineages with independent age sequences and independent identity draws at every new maternal arrival, continuing independently even after their identities would coincide. Couple the real genealogies to these ghosts up to their first

common maternal individual; afterward follow one common maternal continuation. Conditional on no earlier collision, exploration of distinct vertices exposes independent previously unused edge variables, so the construction has the law of the original pedigree.

For ghost $i \in \{1, 2\}$ write $A_{i,1}, A_{i,2}, \ldots$ and define

$$S_{i,0} = 0, \quad S_{i,k} = \sum_{\ell=1}^{k} A_{i,\ell}, \quad R_i(m) = \max\{k \geq 0 : S_{i,k} \leq m\}. \tag{2}$$

The calendar first-crossing generation depth is

$$G_i(m) = \min\{k \geq 0 : S_{i,k} \geq m\}, \qquad m \geq 1. \tag{3}$$

As the $S_{i,k}$ are integers with strictly positive increments, $|G_i(m) - R_i(m)| \leq 1$. For the real paths, let $D_n^{\text{real}}(m)$ denote the difference of their first-crossing depths to the same calendar boundary $-m$.

Let T_n be the earliest positive calendar date, measured backward, at which the ghost paths reach the *same identified mother*. Define $T_n = \infty$ if they never meet. Under the coupling, T_n is precisely the first real maternal coalescence time. Whenever both ghosts renew at date t, their maternal identities there are independent uniforms on the same N_n-element set, and hence coincide with probability $1/N_n$; these label comparisons are independent over distinct dates conditional on the two age sequences.

Remark 2.1 (Biological and statistical meaning). *The ages* 1 *and* 2 *are* calendar units in a deliberately simplified mathematical population*, not claims about human childbirth ages. The model has positive nonzero maternal-age variance, genuine many-to-one maternal identities, and real coalescence. It is a proof-of-principle model, not a historical fit.*

3 Renewal occupation and its Gaussian fluctuation

Set

$$\mu = \mathbb{E}[A] = 2 - p, \qquad \sigma^2 = \text{Var}(A) = p(1-p), \qquad r = \mu^{-1}. \tag{4}$$

Introduce a residual-time Markov chain $X_i(t) \in \{0, 1\}$, initially $X_i(0) = 0$, with transition matrix

$$P = \begin{pmatrix} p & 1-p \\ 1 & 0 \end{pmatrix}. \tag{5}$$

The state 0 at $t \geq 1$ signifies a renewal (maternal birth date). Thus

$$R_i(m) = \sum_{t=1}^{m} \mathbf{1}_{\{X_i(t)=0\}}.$$

The chain is irreducible and aperiodic for $0 < p < 1$. Its stationary law is $\pi = (r, 1-r)$.

Lemma 3.1 (Uniform joint occupation law). *For every fixed $S < \infty$, with $I_t = \mathbf{1}_{\{X_1(t)=X_2(t)=0\}}$ and $M_n(u) = \sum_{1 \leq t \leq \lfloor nu \rfloor} I_t$,*

$$\sup_{0 \leq u \leq S} \left| \frac{M_n(u)}{n} - r^2 u \right| \xrightarrow{\mathbb{P}} 0. \tag{6}$$

Proof. The product chain (X_1, X_2) is an irreducible, aperiodic, finite-state Markov chain, with invariant law $\pi \otimes \pi$. The strong law applied to $f(x, y) = \mathbf{1}_{\{(x,y)=(0,0)\}}$ gives $m^{-1} \sum_{t \leq m} I_t \to r^2$ almost surely. Pointwise convergence at finitely many mesh points, monotonicity of $M_n(u)$, and continuity of $r^2 u$ imply uniform convergence on compact time intervals: approximate any u between mesh points by its two neighbours, then let the mesh size vanish. The arbitrary initial state does not affect the ergodic law. □

Lemma 3.2 (Two-lineage functional central limit theorem). *For $i = 1, 2$, set*

$$Y_{i,n}(u) = n^{-1/2}\{R_i(\lfloor nu \rfloor) - nu/\mu\}.$$

In $D([0, S], \mathbb{R}^2)$,

$$(Y_{1,n}, Y_{2,n}) \Rightarrow \frac{\sigma}{\mu^{3/2}}(B_1, B_2), \tag{7}$$

where B_1, B_2 are independent standard Brownian motions (their signs may be absorbed into their definitions).

Proof. Write $f(x) = \mathbf{1}_{\{x=0\}} - r$. The Poisson equation $h - Ph = f$ has the explicit solution $h(0) = r$, $h(1) = 0$: at state 0, $h(0) - (Ph)(0) = r(1-p) = 1 - r$, and at state 1 it equals $-r$. Consequently

$$\sum_{t=0}^{m-1} f(X_i(t)) = h(X_i(0)) - h(X_i(m)) + \sum_{t=1}^{m}\{h(X_i(t)) - (Ph)(X_i(t-1))\}. \tag{8}$$

The last sum is a martingale with bounded differences. Its predictable quadratic variation per unit time tends by the ergodic theorem to

$$\pi(0)\,\mathrm{Var}(h(X(t+1)) \mid X(t) = 0) = r\,\frac{p(1-p)}{\mu^2} = \frac{\sigma^2}{\mu^3}. \tag{9}$$

Indeed, the transition out of state 1 is deterministic and gives zero conditional variance. The bounded-increment martingale functional CLT applies; the boundary term in (8) is $O(1)$, uniformly over time. Since the two chains are independent, their martingales have independent limiting Brownian motions. Shifting the sum from $t = 0$ to $t = 1$ and replacing $\lfloor nu \rfloor r$ by nur change the numerator by at most a constant. □

Proposition 3.3 (Quantitative renewal occupation bound). *There exists a constant $C_{p,S} < \infty$ such that, uniformly over the four possible initial states of the product chain,*

$$\mathbb{E}\left[\sup_{0 \le u \le S}\left|\frac{M_n(u)}{n} - r^2 u\right|\right] \le \frac{C_{p,S}}{\sqrt{n}}. \tag{10}$$

The same bound, with the same constant, applies following any non-anticipative restart at which the future transitions form a fresh copy of the product chain conditional on its current state.

Proof. For the four-state transition matrix $Q = P \otimes P$, the centred function $f(x, y) = \mathbf{1}_{\{(x,y)=(0,0)\}} - r^2$ has invariant mean zero. Irreducibility and aperiodicity imply that the Poisson equation $h - Qh = f$ has a bounded solution h on the finite state space. The same martingale decomposition as in (8) gives

$$\sum_{t=0}^{m-1} f(Y_t) = h(Y_0) - h(Y_m) + \mathcal{M}_m,$$

where $|\mathcal{M}_m - \mathcal{M}_{m-1}| \le 2\|h\|_\infty$ and $\mathcal{M}$ is a martingale. Doob's L^2 maximal inequality yields

$$\mathbb{E}\max_{m \le \lfloor nS \rfloor} |\mathcal{M}_m| \le 2\sqrt{\mathbb{E}|\mathcal{M}_{\lfloor nS \rfloor}|^2} \le 4\|h\|_\infty \sqrt{nS}.$$

The bounded endpoint terms, shifting the time index, and rounding nu add $O(1)$ uniformly. Dividing by n proves (10). After a non-anticipative restart, condition on the state at restart and use the identical uniform bound; it is crucial that the future transitions have not been exposed by the enlarged conditioning. □

4 Joint collision limit and the stopped real depth

At each date $t \geq 1$ with $I_t = 1$, let $C_{n,t}$ be the indicator that the two independently chosen mother labels coincide. The collision point process is

$$\Xi_n = \sum_{t \geq 1} I_t C_{n,t}\, \delta_{t/n}. \tag{11}$$

Its first point equals T_n/n. Write $\tau_n = T_n/n$. For later use set $Z_n = Y_{1,n} - Y_{2,n}$ and, for $u \geq 0$, $Z_n^{\mathrm{stop}}(u) = Z_n(u \wedge \tau_n)$, interpreting $u \wedge \infty = u$.

Theorem 4.1 (Constructive joint depth–collision limit). *Under the fully specified maternal-identity model of Section 2, for every $S < \infty$,*

$$\big(Y_{1,n}, Y_{2,n}, \Xi_n\big) \Rightarrow \left(\frac{\sigma}{\mu^{3/2}} B_1, \frac{\sigma}{\mu^{3/2}} B_2, \Xi\right), \tag{12}$$

in $D([0,S],\mathbb{R}^2)$ times the space of locally finite point measures on $[0,S]$. Here Ξ is a homogeneous Poisson point process with intensity

$$\lambda = \frac{1}{\nu\mu^2} = \frac{1}{\nu(2-p)^2}, \tag{13}$$

and is independent of (B_1, B_2). Its first point τ has the exponential survival law $\mathbb{P}(\tau > u) = e^{-\lambda u}$. Moreover, in $D([0,S],\mathbb{R})$, jointly with τ_n,

$$\left(\frac{D_n^{\mathrm{real}}(\lfloor n\cdot \rfloor)}{\sqrt{n}}, \tau_n\right) \Rightarrow \big(Z(\cdot \wedge \tau), \tau\big), \tag{14}$$

where $Z = \frac{\sigma}{\mu^{3/2}}(B_1 - B_2)$ has variance rate $2\sigma^2/\mu^3$ and is independent of τ. In particular the real depth difference *freezes at the first shared mother in the limit.*

Proof. Conditional on the two complete ghost age sequences, the variables $C_{n,t}$ for the matching renewal dates are independent Bernoulli variables with parameter $1/N_n$. Thus, for continuous $g \geq 0$ supported in $[0,S]$,

$$L_n(g) := \mathbb{E}[e^{-\langle g, \Xi_n\rangle} \mid X_1, X_2] = \prod_{1 \leq t \leq nS} \left(1 - \frac{I_t}{N_n}(1 - e^{-g(t/n)})\right). \tag{15}$$

Since $I_t \leq 1$, the logarithm differs from $-N_n^{-1}\sum_{t \leq nS} I_t(1 - e^{-g(t/n)})$ by at most $Cn/N_n^2 = o(1)$, uniformly in the age trajectories. Lemma 3.1 and a Stieltjes/Riemann-sum approximation yield, in probability,

$$L_n(g) \longrightarrow \exp\left\{-\lambda \int_0^S (1 - e^{-g(u)})\, du\right\} =: L(g). \tag{16}$$

The convergence holds in L^1, as $0 \leq L_n \leq 1$. For any bounded continuous functional F of the two age-chain trajectories,

$$\begin{aligned}\mathbb{E}[F(Y_{1,n}, Y_{2,n}) e^{-\langle g, \Xi_n\rangle}] &= \mathbb{E}[F(Y_{1,n}, Y_{2,n}) L_n(g)] \\ &= L(g)\mathbb{E}[F(Y_{1,n}, Y_{2,n})] + o(1).\end{aligned}$$

Together with Lemma 3.2, this factorization identifies the *joint* limit as the product of the Brownian law and the Poisson law. The standard Laplace-functional convergence criterion for locally finite point measures completes (12). Notice that independent marginals alone would not have been enough.

We also control the origin. Taking g approximations to the indicator of $(0, u]$, or using directly the conditional survival identity

$$\mathbb{P}(\tau_n > u \mid X_1, X_2) = (1 - 1/N_n)^{M_n(u)}, \tag{17}$$

we obtain $\mathbb{P}(\tau_n > u \mid X_1, X_2) \to e^{-\lambda u}$ in probability for each $u > 0$. Thus no hidden collision mass can accumulate at zero. The independent limit has no atoms at any deterministic positive time; first-point convergence is valid on compact intervals with the cemetery value ∞ when there is no point. By consistency in S, $\tau_n \Rightarrow \tau$ on $[0, \infty]$.

Up to the first shared mother, the real path and its ghost are identical. From that shared vertex onward the real ancestral paths are identical. Hence the difference of their completed-renewal counts is *exactly* the ghost count difference stopped at T_n for every integer date. The first-crossing depth differs from the completed-renewal count by at most one per lineage, so

$$\sup_{0 \le u \le S} \left| \frac{D_n^{\mathrm{real}}(\lfloor nu \rfloor)}{\sqrt{n}} - Z_n(u \wedge \tau_n) \right| \le \frac{2}{\sqrt{n}}, \tag{18}$$

with the convention $D_n^{\mathrm{real}}(0) = 0$. Now use the joint convergence, the almost sure continuity of Brownian paths, and the continuous mapping theorem for stopping at a time with no deterministic atoms. This proves (14). □

Remark 4.2 (What the theorem does not imply). *Theorem 4.1 is a genuine, non-degenerate joint-limit result, but for a* single homogeneous maternal pool *with bounded ages* $1, 2$*, exchangeable uniform mother selection, and two descendants. It does not establish the spatial Cox mixture, Palm restart with unequal weights,* $k_n \uparrow \infty$*, dense cohort handoff, or historical identification of the full M0–M6c programme.*

Corollary 4.3 (Deterministically weighted maternal identities). *Replace uniform mother selection by a deterministic probability vector* $w^{(n)} = (w_v^{(n)})_{v=1}^{N_n}$*, the same at every birth date, independent of ages and of all other edge choices. Suppose*

$$c_n = \sum_v (w_v^{(n)})^2, \qquad nc_n \longrightarrow c \in (0, \infty).$$

Then Theorem 4.1 holds with collision intensity $\lambda = cr^2 = c/\mu^2$*, with the same Gaussian variance. The selected mother's identity at the first collision has conditional law* $(w_v^{(n)})^2/c_n$*, not the original* $w^{(n)}$*.*

Proof. Conditional on the independent ghost age sequences, each matching-date pair of mother labels collides with probability exactly c_n, independently over dates. Replace $1/N_n$ by c_n in (15)–(17); the logarithmic remainder is $O(nc_n^2) = o(1)$ and the occupation limit is unchanged. The Palm identity follows from Bayes' rule. After a merger the selected mother's own age remains an independent sample of (1), because edge ages are attached to distinct vertices independently of their being selected as a parent. Thus the real-ghost first-merger coupling and stopping proof are unchanged. □

4.1 A quantitative conditional Poisson approximation

Proposition 4.4 (Quantitative conditional collision-Laplace bound). *Fix* $S < \infty$ *and let* $g \ge 0$ *be Lipschitz on* $[0, S]$*. Put* $h = 1 - e^{-g}$*,* $L_n(g) = \mathbb{E}[e^{-\langle g, \Xi_n \rangle} \mid X_1, X_2]$ *(as defined below), and* $L(g) = \exp\{-r^2\nu^{-1} \int_0^S h(u)\, du\}$*. Then for a constant* $C = C(p, \nu, S, g)$ *and all sufficiently large* n*,*

$$\mathbb{E}\,|L_n(g) - L(g)| \le \frac{C}{\sqrt{n}}. \tag{19}$$

More explicitly, if $\delta_n = \sup_{0\le u\le S} |M_n(u)/n - r^2 u|$,

$$\mathbb{E}|L_n(g) - L(g)| \le \frac{2\lfloor nS\rfloor}{N_n^2} + \frac{n}{N_n}\big(\|h\|_\infty + S\,\mathrm{Lip}(h)\big)\mathbb{E}\delta_n \tag{20}$$

$$+ r^2 S\|h\|_\infty \left|\frac{n}{N_n} - \frac{1}{\nu}\right|. \tag{21}$$

This is a bound on the conditional Laplace functional *in the toy pedigree, not a finite-n total-variation bound for the full ancestral tree.*

Proof. Conditional on both ghost age sequences, the collision indicators at common renewal dates are independent Bernoulli$(1/N_n)$ trials, so $L_n(g) = \prod_{t\le nS}(1 - I_t h(t/n)/N_n)$. For $0 \le x \le 1/2$, $|\log(1-x) + x| \le 2x^2$. For $N_n \ge 2$ this yields

$$\left|\log L_n(g) + \frac{1}{N_n}\sum_{t\le nS} I_t h(t/n)\right| \le \frac{2\lfloor nS\rfloor}{N_n^2}.$$

Stieltjes summation by parts, with $M_n(u)/n$ the empirical occupation distribution, gives (interpreting the derivative of the Lipschitz function h almost everywhere)

$$\left|\frac{1}{n}\sum_{t\le nS} I_t h(t/n) - r^2\int_0^S h(u)\,du\right| \le \big(\|h\|_\infty + S\,\mathrm{Lip}(h)\big)\delta_n.$$

The map $x \mapsto e^{-x}$ is 1-Lipschitz on $[0,\infty)$. The triangle inequality proves (21). Proposition 3.3 bounds $\mathbb{E}\delta_n$ by $C_{p,S}n^{-1/2}$; $N_n = \lfloor \nu n\rfloor$ implies $n/N_n = \nu^{-1} + O(n^{-1})$. These bounds give (19). □

Remark 4.5 (Conditional, not unconditional, information). *The L^1 estimate is stronger than convergence of a single averaged Laplace transform: it controls the random conditional transform given both age histories. This is exactly the input used to factorize the joint Gaussian and collision limits; it is still specific to a common pool with exogenous identity draws.*

5 A finite-regime historical environment and Cox mixing

The homogeneous limit extends rigorously to a class of exogenous environments with finitely many changes. This provides a genuinely time-inhomogeneous model with a random collision intensity, but it does *not* introduce endogenous geographic migrations or weighted post-Palm class transitions.

Let J take values in a fixed finite set $\mathcal{J}$, with probabilities $\alpha_j > 0$, independently of the genealogical edge randomness conditional on its value. For each j fix a deterministic partition

$$0 = s_0 < s_1 < \cdots < s_R = S,$$

and constants $p_{j,r} \in [\eta, 1-\eta]$, $\nu_{j,r} \in [\eta, \eta^{-1}]$ for some $\eta > 0$. The cohort at calendar date $-t$ has $N_{n,j}(t) = \lfloor n\nu_{j,r}\rfloor$ mother identities if $t/n \in [s_{r-1}, s_r)$; the edge age of an individual born at date $-t$ takes values $1, 2$ with probability $p_{j,r}$ for age 1. Each child selects uniformly from the corresponding mother-date cohort; all vertex edge choices remain independent conditional on J. At the finitely many interface dates we choose a fixed one-sided convention, immaterial to limits.

Define piecewise-constant functions, conditional on $J = j$,

$$r_j(u) = \frac{1}{2 - p_{j,r}}, \qquad v_j(u) = \frac{p_{j,r}(1-p_{j,r})}{(2-p_{j,r})^3}, \qquad \lambda_j(u) = \frac{1}{\nu_{j,r}(2-p_{j,r})^2}$$

for $u \in [s_{r-1}, s_r)$. Write $H_j(u) = \int_0^u r_j(s)\,ds$ and $\Lambda_j(u) = \int_0^u \lambda_j(s)\,ds$.

Theorem 5.1 (Finite-regime Gaussian–Cox limit with stopping). *For the finite-regime model above, conditional on $J = j$,*

$$\left(\frac{R_1(\lfloor n\cdot\rfloor) - nH_j(\cdot)}{\sqrt{n}}, \frac{R_2(\lfloor n\cdot\rfloor) - nH_j(\cdot)}{\sqrt{n}}, \Xi_n\right) \Rightarrow (B_{1,j}, B_{2,j}, \Xi_j), \tag{22}$$

where $B_{i,j}(u) = \int_0^u \sqrt{v_j(s)}\, dW_i(s)$ are conditionally independent Gaussian martingales, and Ξ_j is a Poisson point process of deterministic intensity $\lambda_j(u)\, du$, conditionally independent of both Gaussian processes. Jointly with J, the unconditioned limit Ξ is a Cox process directed by the random *intensity $\lambda_J(u)\, du$. The first-coalescence time Θ satisfies*

$$\mathbb{P}(\Theta > u \mid J = j) = e^{-\Lambda_j(u)}.$$

The actual genealogical depth difference*, divided by $\sqrt{n}$, converges conditionally to*

$$\big(B_{1,j} - B_{2,j}\big)(\cdot \wedge \Theta), \tag{23}$$

with a stopping time independent of the Gaussian paths conditional on J. Conditional independence need not survive mixing over the common environment J.

Proof. Fix an environment j, so all epoch boundaries and parameters are now deterministic. Within each open epoch, outside a bounded number of calendar steps next to its boundaries (maternal increments are at most 2), each ghost residual chain uses the homogeneous transition matrix $P(p_{j,r})$. Its Poisson-equation decomposition (Lemma 3.2) gives a bounded martingale increment sequence with mean reward $r_{j,r}$ and limiting quadratic variation per unit time $v_{j,r}$. Summing the decompositions across the fixed number R of epochs adds at most $O(R)$ boundary terms, which vanish after $\sqrt{n}$ scaling. The bracket convergence follows from the finite-state ergodic theorem in each epoch of macroscopic positive length. The martingale functional CLT for bounded triangular arrays then yields the two Gaussian martingales with the displayed deterministic time changes. Their independent driving noises follow from the independence of the two ghost edge streams conditional on j.

The same epochwise argument for the product residual chain gives, uniformly in $u \le S$,

$$\frac{1}{n}\sum_{t\le nu} \mathbf{1}_{\{X_1(t)=X_2(t)=0\}} - \int_0^u r_j(s)^2\, ds \xrightarrow{\mathbb{P}} 0$$

with finite interfaces contributing $O(R/n)$. More generally, weighting the common-renewal sum by the reciprocal of the mother-cohort size at each date gives the limiting cumulative collision compensator

$$\sum_{t\le nu} \frac{\mathbf{1}_{\{X_1(t)=X_2(t)=0\}}}{N_{n,j}(t)} \xrightarrow{\mathbb{P}} \int_0^u \frac{r_j(s)^2}{\nu_j(s)}\, ds = \Lambda_j(u).$$

Conditional on both complete ghost age paths, the label comparisons remain independent Bernoulli trials, now with success probability $N_{n,j}(t)^{-1}$ at date t. The logarithmic remainder in the conditional Laplace transform is $O(n \sup_t N_{n,j}(t)^{-2}) = o(1)$, and the weighted occupation convergence identifies the conditional point-process Laplace functional of intensity $\lambda_j(u)\, du$. As in Theorem 4.1, the limit is deterministic at each fixed j, so the conditional Laplace factorization proves joint Gaussian–Poisson independence rather than merely marginal convergence. The conditional survival product and the bounded initial-date collision probability exclude a hidden first point at zero.

The exact real-versus-ghost first-meeting coupling remains valid across epoch boundaries because all mothers at a given date have a unique identity and strictly older birth dates. The difference of completed-generation counts is frozen from the first common maternal vertex, and the calendar first-crossing discrepancy is uniformly bounded by 2; the continuous mapping argument proves (23). Finally $\mathcal{J}$ is finite: summing the conditional limits against α_j gives the joint limit with J, and conditional Poisson laws mixed by the shared latent environment define the stated Cox process. □

Remark 5.2 (Precise limit of generality). *The proof uses a* finite *number of exogenous epochs with positive macroscopic lengths, uniformly aperiodic bounded-age kernels, and a common environment chosen before the genealogy. It does not verify the much harder M3b–M3d theory where two ancestral lineages have endogenous and possibly distinct geographic class histories, nor the M4f dense regime.*

5.1 Mixing can restore dependence

The phrase "conditionally independent" requires a precise referent: it is the *un-stopped Gaussian driver* and the collision clock that are independent given J. The stopped process $Z(\cdot \wedge \Theta)$ is generally dependent on Θ, even at fixed J, by construction.

Proposition 5.3 (A covariance diagnostic for a Cox environment). *In Theorem 5.1, write $Z_j = B_{1,j} - B_{2,j}$, so that $V_j(S) = \mathbb{E}[Z_j(S)^2 \mid J = j] = 2\int_0^S v_j(u)\,du$. Then, with $\Lambda_j(S)$ the cumulative collision hazard,*

$$\operatorname{Cov}\big(Z_J(S)^2, \mathbf{1}_{\{\Theta > S\}}\big) = \operatorname{Cov}_J\big(V_J(S), e^{-\Lambda_J(S)}\big). \tag{24}$$

In particular, the two quantities on the left need not be independent after mixing demographic scenarios. If the finite environment consists of two scenarios with probabilities α and $1-\alpha$, the covariance equals

$$\alpha(1-\alpha)\big(V_1(S) - V_2(S)\big)\big(e^{-\Lambda_1(S)} - e^{-\Lambda_2(S)}\big).$$

Proof. Conditional independence in Theorem 5.1 implies $\mathbb{E}[Z_J(S)^2 \mathbf{1}_{\{\Theta>S\}} \mid J = j] = V_j(S)e^{-\Lambda_j(S)}$. Average over J and subtract the product of the unconditional expectations. Expanding the covariance for a two-point mixture gives the last display. □

6 Finite-size calculations and falsifiable checks

The preceding proof permits an exact, non-simulated finite-n check. Order the four product-chain states as $00, 01, 10, 11$; let D_N be diagonal with entry $1 - 1/N$ at 00 and 1 at the other states. Put $Q = P \otimes P$ and let e_{00} be the row distribution concentrated at 00. For integer $m \geq 1$,

$$\mathbb{P}(T_n > m) = e_{00}(QD_{N_n})^m \mathbf{1}. \tag{25}$$

This is exact: at every visited date the two residual states transition, and a shared renewal arrival is killed with probability $1/N_n$. Formula (25) can be independently checked by enumerating all product-chain paths of length m and applying the corresponding survival weights.

Table 1: Exact finite-size survival, $p = 0.6$, $\nu = 1$, $S = 2$; limit $\exp[-2/(1.4)^2]$. Values arise from the 4×4 transfer matrix, not a Monte Carlo study.

$N_n = n$	$\mathbb{P}(T_n/n > 2)$	Absolute error versus limit
25	0.356618	0.003830
50	0.358545	0.001903
100	0.359499	0.000949
250	0.360069	0.000379
500	0.360259	0.000189
1000	0.360353	0.000095
2500	0.360410	0.000038
10000	0.360438	0.000009

These entries are reproduced by the companion Python checks. The unrounded values, numerical tolerances, and independent path enumeration are provided in the package. Agreement is evidence that the finite formula and implementation are consistent; *the proof is the preceding argument*, not the table.

Table 2: Exact two-epoch conditional survival at $S = 2$ ($n = 2500$), compared with the finite-regime Cox limits. Epochs have unit macroscopic duration. The unconditional mixture differs from the survival obtained by averaging its hazards before exponentiating.

Scenario	(p_1, ν_1)	(p_2, ν_2)	Exact	Cox limit
$j = 0$	$(0.35, 1)$	$(0.70, 2)$	0.515195	0.515217
$j = 1$	$(0.80, 2)$	$(0.30, 1)$	0.499941	0.499952

With equal scenario weights, the limiting mixture survival equals 0.507584, while exponentiating the average cumulative hazard gives 0.507527. The numerical gap is small for these parameters, but the two operations are mathematically distinct, by Jensen's inequality. The results are reproduced in `tests/finite_regime_cox_check.py`.

7 Identity bias, Palm selection, and simultaneous mergers

When the labels of two mother selections are not uniform but have probabilities $w_1, \dots, w_N$ on one maternal pool, two independent draws collide with exact probability

$$c(w) = \sum_{v=1}^{N} w_v^2. \tag{26}$$

If a collision occurs, the common mother V has the *Palm-biased* identity law

$$\mathbb{P}(V = v \mid \text{collision}, w) = \frac{w_v^2}{\sum_u w_u^2}. \tag{27}$$

For two different daughter types with weights w and z, replace w_v^2 by $w_v z_v$, and normalize by $\sum_u w_u z_u$. Formula (27) is an elementary Bayes identity, *not* by itself a theorem of uniform post-coalescence restart.

Proposition 7.1 (Finite-date multi-merger union bound). *Suppose that, at each of at most $\lfloor nS \rfloor$ past dates, no more than k_n distinct active ancestral lines choose independent uniform mothers among $N_n \geq cn$ identities. The probability of at least one date exhibiting either a triple choice of the same mother, or two disjoint simultaneous binary collisions, is bounded by*

$$\lfloor nS \rfloor \left\{ \frac{\binom{k_n}{3}}{N_n^2} + \frac{3\binom{k_n}{4}}{N_n^2} \right\} = O_S\left(\frac{k_n^3 + k_n^4}{n} \right). \tag{28}$$

In particular it tends to zero if $k_n = o(n^{1/4})$.

Proof. For any three specific distinct lines, simultaneous equality of three independent uniform parent labels has probability N_n^{-2}. For any four distinct lines there are three pairings into two disjoint pairs, each occurring with probability N_n^{-2} (whether or not the two pair labels are distinct only enlarges the bound). Apply a union bound over all subsets, pairings, and dates. The conditional version is valid for non-anticipative selection of active lineages because parent labels at that date have not yet been revealed. □

Remark 7.2. *The bound proves the absence of these* particular simultaneous events *in one simple model, not an approximation of the complete many-lineage genealogy by a Kingman coalescent. It does not derive the uniform pair-lifetime bounds or the dense–sparse handoff required by M4e–M5c.*

8 Descendant-weighted dependence and cohort disparity

Let $N \geq 2$ contemporary descendants be divided into m nonempty ancestral blocks $B_1, \ldots, B_m$, with sizes $s_j \geq 1$, $\sum s_j = N$, and descendant weights $w_j = s_j/N$. Assign the same depth mark a_j to every descendant in block B_j. The block measure is $\Lambda = \sum_{j=1}^{m} w_j \delta_{a_j}$. Define

$$\mathsf{D}(\Lambda) = \iint |x - y| \, d\Lambda(x) \, d\Lambda(y), \quad \mathsf{R}(\Lambda) = \max \operatorname{supp} \Lambda - \min \operatorname{supp} \Lambda. \tag{29}$$

These are with-replacement pair disparity and cohort range, respectively. Their empirical interpretation depends on the actual ancestral partition and cut definition.

Proposition 8.1 (Exact dispersion under binary block marks). *Fix $a < b$ and $d = b - a > 0$. If the mark of block j is $a_j = a + dZ_j$ with $Z_j \in \{0, 1\}$, and $K = \sum_j w_j Z_j$, then, for every deterministic assignment,*

$$\mathsf{D}(\Lambda) = 2dK(1 - K), \qquad \mathsf{R}(\Lambda) = d \, \mathbf{1}_{\{0 < K < 1\}}. \tag{30}$$

Consequently the exact maximum of D over all assignments is $2d \max_{t \in \mathcal{S}} t(1-t)$, where $\mathcal{S} = \{\sum_{j \in J} w_j : J \subseteq \{1, \ldots, m\}\}$. If instead the Z_j are independent Bernoulli(q) and $H = \sum_j w_j^2$, then

$$\mathbb{E}\mathsf{D}(\Lambda) = 2dq(1 - q)(1 - H), \qquad \mathbb{E}\mathsf{R}(\Lambda) = d\{1 - q^m - (1 - q)^m\}. \tag{31}$$

Proof. The total descendant weight at b is K and at a is $1 - K$, so the ordered pairs with different marks have combined probability $2K(1 - K)$. The range is d exactly if both marks appear. Deterministically $K \in \mathcal{S}$, and every subset of blocks is attainable. For independent Bernoulli marks, $\mathbb{E}K = q$ and $\operatorname{Var}(K) = q(1 - q)H$. Hence $\mathbb{E}[K(1 - K)] = q(1 - q)(1 - H)$. All blocks have the same mark with probability $q^m + (1 - q)^m$, giving the second expectation. □

Proposition 8.2 (Exact role of ancestral-mark dependence). *Let the ancestral partition and weights w_j be fixed, and let $Z_j \in \{0, 1\}$ have arbitrary joint law, with $q_j = \mathbb{E}Z_j$. Assign the marks $a_j = a + dZ_j$ with $d > 0$. Then*

$$\mathbb{E}\mathsf{D}(\Lambda) = 2d \sum_{1 \leq i < j \leq m} w_i w_j \, \mathbb{P}(Z_i \neq Z_j) \tag{32}$$

$$= 2d \sum_{i<j} w_i w_j \Big(q_i + q_j - 2q_i q_j - 2 \operatorname{Cov}(Z_i, Z_j) \Big). \tag{33}$$

If all $q_j = q$, this becomes

$$\mathbb{E}\mathsf{D}(\Lambda) = 2dq(1 - q)(1 - H) - 4d \sum_{i<j} w_i w_j \operatorname{Cov}(Z_i, Z_j), \qquad H = \sum_i w_i^2. \tag{34}$$

No independence or exchangeability is assumed. For fixed marginals, Fréchet–Hoeffding constraints imply the pairwise bounds

$$|q_i - q_j| \leq \mathbb{P}(Z_i \neq Z_j) \leq \min\{q_i + q_j, \, 2 - q_i - q_j\}.$$

The resulting sum gives valid lower and upper bounds for expected cohort disparity, but the pairwise extrema need not be simultaneously *attainable for all pairs in a multi-block law.*

Proof. For deterministic binary marks, $\mathsf{D}(\Lambda) = \sum_{i,j} w_i w_j d|Z_i - Z_j| = 2d \sum_{i<j} w_i w_j \mathbf{1}_{\{Z_i \neq Z_j\}}$. Take expectations, then use $\mathbb{E}[Z_i Z_j] = q_i q_j + \operatorname{Cov}(Z_i, Z_j)$. For equal q, $\sum_{i<j} w_i w_j = (1 - H)/2$, giving (34). Finally $\mathbb{P}(Z_i \neq Z_j) = q_i + q_j - 2\mathbb{P}(Z_i = 1, Z_j = 1)$, with $\max(0, q_i + q_j - 1) \leq \mathbb{P}(Z_i = 1, Z_j = 1) \leq \min(q_i, q_j)$. Substituting these Fréchet bounds proves the last display. □

Remark 8.3 (What the covariance identity does and does not identify). *Equation* (34) *makes the missing dependence information explicit: the one-block marginals q_j alone do not determine mean dispersion. Positive mark covariance reduces the expected disparity relative to independent marking with the same marginals; negative covariance raises it. The result is an algebraic identity for assigned block marks, not an estimate of those covariances from historical data or a consequence of coalescence without an additional mark model.*

Remark 8.4 (A measurement warning). *The factor H is the probability that two leaves independently drawn* with replacement *belong to the same block of the particular ancestral partition. It is not a universal historical coalescence probability. The formula for $\mathbb{E}\mathsf{D}$ assumes independent block marks; coalescent identities alone do not imply independent marking.*

Example 8.5 (Identical individual marginals, different joint disparity). *For $N = 4$, choose a fair common coin and assign all four descendants depth a or all depth b. In a second law, choose uniformly a two-person subset to receive b and give the other two a. Every person's marginal law is $\frac{1}{2}(\delta_a + \delta_b)$ in both constructions. Under the common coin the realized cohort disparity and range vanish. Under the balanced law the with-replacement disparity equals $d/2$, the without-replacement disparity equals $2d/3$, and the range equals d. This proves nonidentifiability from marginals in a simple exact sense; it is* not *a newly calibrated genealogical model.*

9 Growing samples and whole-cohort control in the explicit model

This section adds a result for the *entire* date-zero population to the exact homogeneous maternal-identity model of Section 2. It does not assert that the spatial dense–sparse handoff of the general M4–M5 theory has been proved. The first contribution is an elementary but quantitative simultaneous estimate, not a new coalescent limit. The second connects the non-degenerate pairwise stopped limit to an exact *expected whole-cohort* statistic without incorrectly assuming that members of the cohort are independent.

For every date-zero woman $i \in \{1, \dots, N_n\}$, let $G_{i,n}(m)$ be her actual, shared-pedigree first-crossing depth at integer calendar cut $-m$, with $G_{i,n}(0) = 0$. Put

$$\overline{\Delta}_n(m) = \frac{1}{N_n(N_n-1)} \sum_{i \neq j} |G_{i,n}(m) - G_{j,n}(m)|, \tag{35}$$

$$\mathcal{R}_n(m) = \max_{1 \le i \le N_n} G_{i,n}(m) - \min_{1 \le i \le N_n} G_{i,n}(m). \tag{36}$$

Both are *raw integer-generation* observables and neither should be identified with the M_4 structural block variables D_n^A, R_n^A.

Lemma 9.1 (A marginal iid spine survives shared ancestors). *For each fixed date-zero individual i, the successive maternal-age variables encountered along her* actual *ancestral path are iid with $\mathbb{P}(A = 1) = p$ and $\mathbb{P}(A = 2) = 1 - p$, despite the dependence between* different *individuals' paths. Moreover, for every $K \ge 1$ and $x > 0$, writing $H_{i,k} = \sum_{\ell=1}^{k} (A_{i,\ell} - \mu)$,*

$$\mathbb{P}\left(\max_{0 \le k \le K} |H_{i,k}| \ge x \right) \le 2 \exp(-2x^2/K). \tag{37}$$

Proof. Following one ancestral path, birth dates strictly decrease and therefore the visited vertices are distinct. The choice of an ancestral vertex is made by the daughter without consulting that ancestor's as-yet unrevealed outgoing maternal age. Thus, conditionally on the exploration history before that new age is revealed, its law is the original Bernoulli law; iteration proves the iid assertion. For the centred increments, $\mathbb{E}e^{\theta(A-\mu)} \le e^{\theta^2/8}$ by Hoeffding's lemma, since $A \in [1, 2]$. The exponential supermartingale and Doob's maximal inequality yield $\mathbb{P}(\max_{k \le K} H_{i,k} \ge x) \le \inf_{\theta > 0} \exp(-\theta x + K\theta^2/8) = e^{-2x^2/K}$.

Apply the argument to $-H_{i,k}$ and take the union. No independence between maternal paths of different leaves is used. □

Theorem 9.2 (Simultaneous full-cohort depth and range bound). *Fix $S > 0$, let $K_n = \lfloor nS \rfloor + 1$, and define for $\delta \in (0,1)$*

$$x_{n,\delta} = \sqrt{\frac{K_n}{2} \log\left(\frac{2N_n}{\delta}\right)}. \tag{38}$$

Then, with probability at least $1-\delta$, simultaneously *for all $i \le N_n$ and integers $0 \le m \le \lfloor nS \rfloor$,*

$$\left| G_{i,n}(m) - \frac{m}{\mu} \right| \le \frac{x_{n,\delta}+1}{\mu}. \tag{39}$$

Consequently, on that same event,

$$\sup_{0 \le m \le \lfloor nS \rfloor} \mathcal{R}_n(m) \le \frac{2(x_{n,\delta}+1)}{\mu}, \qquad \sup_m \overline{\Delta}_n(m) \le \frac{2(x_{n,\delta}+1)}{\mu}. \tag{40}$$

In particular, for $N_n \sim \nu n$, both quantities divided by n converge to zero in probability, and the same holds for the maximum individual deviation $\max_{i,m} |G_{i,n}(m)/n - m/(n\mu)|$. The explicit finite-$n$ expectation bound

$$\mathbb{E}\left[\sup_{m \le nS} \mathcal{R}_n(m) \right] \le \frac{2}{\mu}\left(1 + a_n + \frac{K_n}{4a_n}\right), \quad a_n = \sqrt{\frac{K_n}{2}\log(2N_n)}, \tag{41}$$

also holds. In particular the upper scale is $O(\sqrt{n \log n})$ when $N_n \asymp n$; this is not *a claim that this upper rate is sharp.*

Proof. For each actual maternal path and integer $m \ge 1$, let $k = G_{i,n}(m)$. Since maternal ages lie in $\{1,2\}$, $k \le m \le K_n$ and $m \le S_{i,k} \le m+1$. Thus $|\mu k - m| \le |H_{i,k}| + 1$. For $m = 0$ the same inequality holds trivially. Apply Lemma 9.1 to the N_n lines, use a union bound *without* any independence assumption between their histories, and substitute $x_{n,\delta}$: the failure probability is at most $2N_n e^{-2x_{n,\delta}^2/K_n} = \delta$. At a fixed m, all depths are within the same radius of the common centre m/μ; this proves the range and average-gap inequalities. Since $\log N_n = o(n)$ when $N_n \asymp n$, division by n proves convergence in probability. Finally, with $W_n = \max_{i,k \le K_n} |H_{i,k}|$, $\mathbb{P}(W_n \ge x) \le \min(1, 2N_n e^{-2x^2/K_n})$. Integrate this tail from a_n upwards and use $\int_a^\infty e^{-2x^2/K_n}\,dx \le K_n e^{-2a^2/K_n}/(4a)$; hence $\mathbb{E}W_n \le a_n + K_n/(4a_n)$. The deterministic bound $\sup_m \mathcal{R}_n(m) \le 2(W_n+1)/\mu$ gives the claim. □

Corollary 9.3 (A bounded-age extension independent of coalescence law). *The simultaneous inequality of Theorem 9.2 continues to hold when the iid maternal ages attached to distinct vertices have arbitrary integer support in $\{a_-, \dots, a_+\}$, with $1 \le a_- < a_+$, mean μ, and their choices are independent of the selected mother identities. Specifically, take $K_n = \lceil nS/a_- \rceil + 1$ and $x = (a_+ - a_-)\sqrt{(K_n/2)\log(2N_n/\delta)}$. With probability at least $1-\delta$, all leaves and cuts $m \le \lfloor nS \rfloor$ satisfy*

$$\left| G_{i,n}(m) - \frac{m}{\mu} \right| \le \frac{x + a_+}{\mu}, \qquad \sup_m \mathcal{R}_n(m) \le \frac{2(x+a_+)}{\mu}.$$

This corollary does not extend the explicit pairwise coefficient *of Theorem 9.4; that coefficient relies on the separately proved two-age stopped limit.*

Proof. The pathwise spine argument is unchanged. Hoeffding's maximal bound for increments of range $a_+ - a_-$ is $2\exp[-2x^2/(K_n(a_+-a_-)^2)]$. For a first crossing at m, the number of renewal steps is at most $\lceil m/a_- \rceil$ and the overshoot is less than a_+. The same union bound and deterministic inversion conclude. □

Theorem 9.4 (A non-degenerate expected disparity for the entire cohort). *Let $S > 0$ be fixed and $N_n = \lfloor \nu n \rfloor \to \infty$. With $\mu = 2-p$, $\sigma^2 = p(1-p)$, $\lambda = (\nu\mu^2)^{-1}$ and $v = 2\sigma^2/\mu^3$, the* actual correlated genealogy *satisfies*

$$\lim_{n\to\infty} \frac{\mathbb{E}\,\overline{\Delta}_n(\lfloor nS \rfloor)}{\sqrt{n}} = \sqrt{\frac{v}{2\lambda}}\,\mathrm{erf}(\sqrt{\lambda S}) = \sqrt{\frac{\nu p(1-p)}{2-p}}\,\mathrm{erf}\left(\frac{\sqrt{S}}{\sqrt{\nu}(2-p)}\right) > 0. \tag{42}$$

*The corresponding independent-renewal (*non-genealogical*) baseline for two depth paths has coefficient*

$$c_{\mathrm{ind}}(S) = \sqrt{\frac{2vS}{\pi}}. \tag{43}$$

For every $S > 0$, the coefficient in (42) is strictly smaller than $c_{\mathrm{ind}}(S)$. Thus this explicit model exhibits a calculable coalescence-induced reduction in expected *whole-cohort pairwise depth disparity on the $\sqrt{n}$ scale. No claim is made about convergence in probability of $\overline{\Delta}_n/\sqrt{n}$ to a deterministic constant.*

Proof. Because the date-zero vertices are exchangeable, for every n and m the expectation of the average over $N_n(N_n-1)$ ordered *distinct* pairs equals the expectation for one fixed distinct pair:

$$\mathbb{E}\,\overline{\Delta}_n(m) = \mathbb{E}\,|G_{1,n}(m) - G_{2,n}(m)|\,. \tag{44}$$

This equality requires neither independence nor concentration of the pair statistics across the cohort. By Theorem 4.1, the right-hand absolute gap divided by $\sqrt{n}$ converges in distribution to $|Z(S \wedge \tau)|$, with Z Brownian of variance rate v, independent of $\tau \sim \mathrm{Exp}(\lambda)$.

To justify passage to expectations, couple the two real lines with independent ghost age sequences up to the first collision. At each calendar cut the real gap differs from the ghost gap stopped at the collision date by at most two. The stopped ghost gap, at any time $\le nS$, is bounded by the sum of the two *maximal* centred renewal deviations; by the martingale maximal L^2 inequality (or Lemma 9.1 integrated in x), each squared maximum has expectation $O(n)$. Therefore the rescaled absolute gaps have uniformly bounded second moments and are uniformly integrable. It follows that the limit of the expectations in (44) is

$$\begin{aligned}
\mathbb{E}|Z(S\wedge\tau)| &= \sqrt{\frac{2v}{\pi}}\,\mathbb{E}\sqrt{S\wedge\tau} \\
&= \sqrt{\frac{2v}{\pi}}\int_0^S \frac{e^{-\lambda t}}{2\sqrt{t}}\,dt \\
&= \sqrt{\frac{v}{2\lambda}}\,\mathrm{erf}(\sqrt{\lambda S}).
\end{aligned}$$

For the independent-renewal benchmark, the pairwise Brownian difference has variance vS without stopping, giving $c_{\mathrm{ind}}(S) = \mathbb{E}|\mathcal{N}(0, vS)|$. Finally $\mathrm{erf}(z) < 2z/\sqrt{\pi}$ for $z > 0$, so the coalescence coefficient is strictly below the independent one. □

Corollary 9.5 (An honest range bracket, not a range limit). *Under Theorem 9.4,*

$$\liminf_{n\to\infty} \frac{\mathbb{E}\,\mathcal{R}_n(\lfloor nS \rfloor)}{\sqrt{n}} \ge \sqrt{\frac{\nu p(1-p)}{2-p}}\,\mathrm{erf}\left(\frac{\sqrt{S}}{\sqrt{\nu}(2-p)}\right), \quad \mathbb{E}\,\mathcal{R}_n(\lfloor nS \rfloor) = O(\sqrt{n\log n}). \tag{45}$$

Proof. The realized range dominates each pairwise absolute difference and hence the average disparity, so take expectations and use Theorems 9.4 and 9.2. □

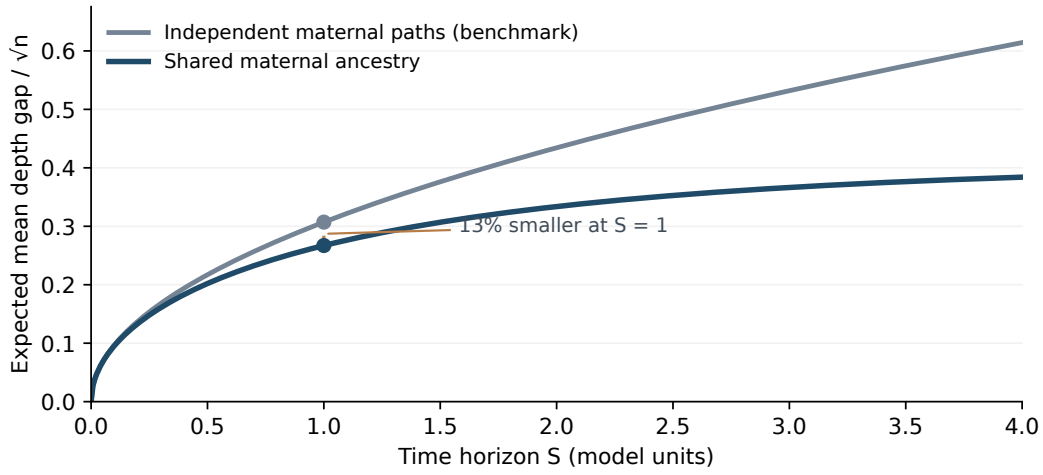


Figure 2: **The theoretical effect of shared ancestors on the expected cohort gap**. The vertical axis is the limiting expected mean pairwise depth difference divided by $\sqrt{n}$, not an observed number of human generations. Curves use the two-age model with $p = 1/2$, $\nu = 1$ and varying model horizon S. At $S = 1$ the coefficients are approximately 0.2671 (shared ancestry) and 0.3071 (independent renewal benchmark). This comparison changes only the dependence mechanism within the stated mathematical model; it does not demonstrate an empirical effect size.

Theorem 9.6 (No simultaneous multiple mergers in a growing sample). *Take $k_n \le N_n$ distinct date-zero leaves in the model, and explore their* actual *maternal genealogy backwards through calendar date $\lfloor nS \rfloor$. At any calendar date the number of active maternal-edge choices is at most k_n. When choices are revealed as batches of fresh independent uniform mother labels for each target birth date, the probability of at least one triple merger or at least two distinct binary mergers at the* same date *is bounded by*

$$\mathbb{P}(\exists \textit{ non-simple collision date} \le \lfloor nS \rfloor) \le \frac{\lfloor nS \rfloor + 2}{N_n^2} \left\{ \binom{k_n}{3} + 3\binom{k_n}{4} \right\}. \tag{46}$$

Consequently, if $N_n \asymp n$ and $k_n = o(n^{1/4})$, all mergers up to macroscopic time S occur as single binary mergers at distinct calendar dates with probability tending to one. This result concerns the combinatorics of coincident parent labels only*; it does not prove a k_n-dimensional functional CLT, a Kingman limit, Palm-restart uniformity, or the dense–sparse handoff of the general M4–M5 theory.*

Proof. Explore backwards in calendar order and defer mother-label assignment to the target birth date. Conditional on the pre-batch exploration history, a list of at most k_n *distinct active children* submits independent choices from the N_n possible mother identities of that date. For any triple of choices, the probability that all three labels agree is N_n^{-2}. For any four choices, there are three possible pairings into two disjoint pairs; each prescribed pairing occurs with probability N_n^{-2}. If at least two distinct binary mergers occur at a date, either a triple equality is present or two disjoint pairs are present. A union bound therefore yields $[\binom{k_n}{3} + 3\binom{k_n}{4}]/N_n^2$ for that date, also when fewer than k_n choices occur. There are at most $\lfloor nS \rfloor + 2$ relevant target birth dates; another union bound proves (46). The bound is $O(k_n^4/n)$ for $N_n \asymp n$. □

Proposition 9.7 (Sampling a realized whole cohort). *Condition on a fixed finite maternal pedigree and its N_n depth marks $g_1, \dots, g_{N_n}$ at a chosen cut. Draw $I_1, \dots, I_k$ independently and uniformly from $\{1, \dots, N_n\}$, allowing repeated indices; put*

$$U_k = \frac{1}{k(k-1)} \sum_{a \ne b} |g_{I_a} - g_{I_b}|, \quad V_{N_n} = \frac{1}{N_n^2} \sum_{i,j} |g_i - g_j|.$$

Writing $R = \max_i g_i - \min_i g_i$, for all $k \geq 2$ and $t > 0$,

$$\mathbb{E}[U_k \mid g_1, \ldots, g_{N_n}] = V_{N_n}, \qquad \mathbb{P}(|U_k - V_{N_n}| > t \mid g_1, \ldots, g_{N_n}) \leq 2\exp\left(-\frac{kt^2}{2R^2}\right). \tag{47}$$

For $R = 0$ the deviation probability is zero by convention. Together with Theorem 9.2, if $k_n/\log n \to \infty$, then

$$(U_{k_n} - V_{N_n})/\sqrt{n} \xrightarrow{\mathbb{P}} 0.$$

This is accuracy of sampling the realized cohort, *not concentration of the random cohort statistic around its ensemble mean.*

Proof. Two different sample positions $a \neq b$ are independent uniforms on the finite index set, so each term has conditional expectation V_{N_n}. Changing a single sampled index affects at most $2(k-1)$ ordered terms, each by at most R, giving a bounded difference of $2R/k$. McDiarmid's inequality proves (47). With probability at least $1 - \delta$, the random range is at most $2(x_{n,\delta} + 1)/\mu$. Substituting $t = \varepsilon\sqrt{n}$ into (47) gives

$$\mathbb{P}(|U_{k_n} - V_{N_n}| > \varepsilon\sqrt{n}) \leq \delta + 2\exp\left(-\frac{k_n\varepsilon^2 n\mu^2}{8(x_{n,\delta}+1)^2}\right).$$

For fixed $\delta > 0$ the exponent tends to $-\infty$ when $k_n/\log n \to \infty$; then let $\delta \downarrow 0$. □

Remark 9.8 (Why the proof cannot be copied into unrestricted M4–M5). *Uniform age control in Theorem 9.2 uses an iid marginal maternal spine with a* single *deterministic mean μ. In a spatial or historically heterogeneous population, distinct descendants can follow macroscopically different maternal-age drifts, and no common centre m/μ need exist. In the general model the block-specific predictable and structural centres, Palm restarts, buffer admissibility, and a dense–sparse approximation must be controlled rather than assumed. Likewise, Theorem 9.4 addresses an ensemble mean and cannot establish self-averaging of the entire cohort's realized dispersion. The two-age law uses dimensionless calendar units and contains no identified human demographic parameter.*

10 A verified deterministic bridge to real-cohort observables

The most delicate part of the full M4–M5 programme is the transition from a random dense genealogy to a sparse list of representative ancestral blocks, and then to a calibrated structural depth mark. The following implication is exact but its *premises* need probabilistic verification in any particular model.

Theorem 10.1 (Uniform coupling transfers to quantiles, disparity and range). *For each n, let $g_{i,n}(u)$ be arbitrary real marks for $N_n \geq 2$ leaves, and let $B_n(i)$ denote their block in a handoff partition, with structural marks $a_{B,n}(u)$. If, on an interval I,*

$$\max_i \sup_{u\in I} |g_{i,n}(u) - a_{B_n(i),n}(u)| \leq \epsilon_n, \tag{48}$$

then, uniformly in $u \in I$, the empirical leaf law $L_{n,u} = N_n^{-1}\sum_i \delta_{g_{i,n}(u)}$ and the descendant-weighted block law $\Lambda_{n,u} = \sum_B (|B|/N_n)\delta_{a_{B,n}(u)}$ obey

$$W_\infty(L_{n,u}, \Lambda_{n,u}) \leq \epsilon_n, \tag{49}$$

$$\sup_{0<v<1} |Q_{L_{n,u}}(v) - Q_{\Lambda_{n,u}}(v)| \leq \epsilon_n, \tag{50}$$

$$|\mathsf{D}(L_{n,u}) - \mathsf{D}(\Lambda_{n,u})| \leq 2\epsilon_n, \tag{51}$$

$$|\mathsf{R}(L_{n,u}) - \mathsf{R}(\Lambda_{n,u})| \leq 2\epsilon_n. \tag{52}$$

The without-replacement pair disparity $\overline{\Delta}_{n,u}$ satisfies

$$|\overline{\Delta}_{n,u} - \mathsf{D}(\Lambda_{n,u})| \leq 2\epsilon_n + \frac{\mathsf{D}(L_{n,u})}{N_n - 1}. \tag{53}$$

Proof. Couple every leaf's mark to its block's mark, using the measure $N_n^{-1}\sum_i \delta_{(g_{i,n}(u),a_{B_n(i),n}(u))}$. The support of this coupling lies within ϵ_n of the diagonal, proving (49). On $\mathbb{R}$, a coupling supported on $|x-y|\le\epsilon_n$ gives the quantile inequality (50). For two independent copies, $||x-y|-|x'-y'||\le |x-x'|+|y-y'|$ proves (51). The maximum and minimum each shift by at most ϵ_n, proving (52). Finally the diagonal terms of the pair sum vanish and $\overline{\Delta}_{n,u}=N_n\mathsf{D}(L_{n,u})/(N_n-1)$, yielding (53). The same coupling works for each u, and the uniform premise gives uniform conclusions. □

11 A mathematically qualified historical compilation interface

The broader M6a–M6c construction is designed to connect documented historical primitives to transition and collision objects. It is a *methodological interface*, not a substitute for the data needed to calibrate it. We state the minimum steps explicitly because otherwise an elegant stochastic limit can be mistaken for a measured statement about human histories.

11.1 From forward maternity flows to a retrospective kernel

Index maternal states by y, child states by x, and possible maternal ages by a. At historical time u, write $\nu_y(u)$ for the appropriate exposed population mass, $f_{y,a}(u)$ for an age-specific fertility factor, and $p^{\rightarrow}_{y,a;x}(u)$ for the destination probability of the child state. When this factorization is justified, its prospective mass is

$$F_{x;y,a}(u)=\nu_y(u)f_{y,a}(u)p^{\rightarrow}_{y,a;x}(u). \tag{54}$$

If the full joint flow F is measured directly, the factorization need not be imposed. Under a positive normalizer, the corresponding retrospective maternal-state distribution is the Bayes inversion

$$q^{\mathrm{geo}}_{x;y,a}(u)=\frac{F_{x;y,a}(u)}{\sum_{z,b}F_{x;z,b}(u)}. \tag{55}$$

Its validity is algebraic. Its historical interpretation requires the flows to be coherent, the population masses and maternal-age exposures to refer to the same population at the same time, and the event definitions to be compatible. A mother–child age gap among contemporaneous household co-residents is *not* automatically an age-specific fertility law.

11.2 From ancestral transitions to collision hazards

For each historical stratum and closed class, a calibrated lifted transition kernel $P_{r,c,u}$ must be stochastic on its specified support. If it has an invariant probability $\pi_{r,c}(u)$, a completion reward $a_{r,c}(u)$ and typed event frequencies $\rho_{\epsilon|r,c}(u)$ can be obtained as stationary expectations. These quantities are not defined merely by naming the geographic labels: an actual communicating-class decomposition and specified interface kernels are required.

The collision enhancement for maternal identity pools is built from a joint reproductive weight law H; it is not generally determined by one-dimensional fertility marginals. In a pair of typed ghost lineages, an intensity has the schematic form

$$\lambda_{r;c,d}(u)=\sum_{\substack{\epsilon,\epsilon':\\ p(\epsilon)=p(\epsilon')}}\rho_{\epsilon|r,c}(u)\rho_{\epsilon'|r,d}(u)\frac{\kappa_{\epsilon,\epsilon'}(u)}{\nu_{p(\epsilon)}(u)}. \tag{56}$$

Equation (56) is a specification of the quantity to compile, not a claim that its inputs are identifiable from the currently available historical records. In particular, demographic scenario values must remain visibly distinguished from measurements.

11.3 The non-equivalence of structural and observed depths

A key boundary concerns *what* the mathematical marks represent. The M4–M5 structural depth $A_{B,n}(u)$ belongs to a handoff block and is formed from model-derived reward integrals. A realized historical depth $G_{i,n}(t)$ is a first crossing of a dated maternal ancestry chain. These quantities do not become equal merely because both are expressed in "number of generations"; one needs an observation or coupling theorem, with a uniform error estimate if extrema of an entire cohort are claimed. Theorem 10.1 spells out the deterministic implication *after* such an error has been proved. The general M4–M5 analysis of that error is not certified by this paper.

Layer	Required mathematical or archival evidence	Status here
Flow inversion	Positive consistent F, documented exposures and time map	Formula only
Lifted kernel	Admissible support, stochasticity, closed classes, interfaces	Contract only
Collision law	Reproductive second moments and identity pools	Model-specific
Handoff marks	Proven uniform comparison to real paths	Not established generally
Human validation	Linked multi-generation maternal pedigrees and held-out tests	Not supplied

12 General historical models: exact scope of the claim

The general programme M0–M6c replaces independent two-age renewals by finite or lifted kernels depending on historical time, reproductive types and geography, with effective maternal pools and reproductive weights. Its mathematical architecture contains (i) local renewal/diffusion arguments, (ii) Cox-type collision clocks conditional on spatial class histories, (iii) post-Palm ancestral restarts, (iv) many-lineage ancestral partitions, (v) a dense–sparse handoff, and (vi) a uniform comparison of individual depths with structural marks.

What has been rigorously established here. Theorems 4.1 and 5.1 give non-degenerate *homogeneous and finite-regime two-lineage* realizations of the joint convergence and stopping mechanism. Proposition 7.1 gives a finite-n union bound for a specified collision event. Proposition 8.1 and Theorem 10.1 are exact and require no asymptotic claims.

What is still an unproved general transfer. The present results do not establish the required spatially conditioned collision compensators, uniform post-merger Palm–Markov dynamics, pair-lifetime estimates for endogenously selected genealogical histories, dense-occupancy descent bounds, simultaneous sparse approximation as the number of lineages increases, non-anticipative handoff under enlarged filtrations, or the full-cohort approximation that would make (48) tend to zero. In the restricted constructions above, finite-regime Cox mixing, static-weight Palm identities and an $O(n^{-1/2})$ finite-state occupation estimate are available; none implies those stronger general statements without further hypotheses and proofs.

M6a specifies historical quantities to be reconstructed; M6b documents sources, gaps and provenance; M6c compiles documented inputs and explicit scenarios into admissible model objects. Passing the frozen M6c contract validator checks the *contract structure*; it does not certify the completeness or accuracy of worldwide historical calibration. The structural marks $A_{B,n}$ are not automatically observed first-crossing depths from a historical linked pedigree. Any such identification needs an additional observation/measurement model.

12.1 A dependency chain that cannot be shortened

For the general full-cohort claim one would need, in logical order: (1) a full micro-level pedigree construction with non-anticipative sampling; (2) joint convergence of depth rewards and first collisions within each persistent class history; (3) an admissible Palm law after each fusion; (4) uniform controls

as the number of lineages grows; (5) a dense-to-sparse handoff valid under its enlarged filtration; and (6) a uniform real-to-mark coupling. The covariance or W_∞ formulas in Sections 8 –10 do not supply steps (3)–(6). Conversely, establishing a single-lineage central limit theorem does not yield any joint dependence claim without at least the collision model in step (2).

This paper closes the complete two-lineage part of this programme for the specified two-age and finite-regime models. It proves the finite deterministic identities that would be used *once* the later steps were available, but does not assert that a general M4–M5 dense-cohort theorem has been proved from the restricted model.

13 Conclusions and outlook

An explicitly constructed maternal genealogy supports a joint Gaussian–Poisson limit for two paths, including coalescent stopping. We derive independence from a conditional Laplace calculation rather than from separate marginal limits. Finite ancestral identity partitions yield exact constraints on pairwise disparity and range, and uniform marked-leaf couplings propagate errors to the extreme and distributional observables of a whole cohort.

The methodological gain is a clean separation of three logically different claims: (1) proving a probability limit within a fully specified mechanism; (2) proving a deterministic inequality conditional on a uniform approximation; and (3) actually verifying that approximation for a dense historical genealogy. The third claim remains open in the broad M4–M5 setting. Extending the constructive proof beyond a homogeneous maternal pool and documenting an empirically identified pedigree are separate challenges. Nothing in this paper gives a numerical estimate of generational dispersion among humans.

From this article to the technical corpus

The companion volume is titled *M0–M6c: Full Technical Corpus*, reader edition of 27 October 2026 (1,406 pages). The routes below are page numbers *in that fixed edition*; they lead to the source mathematics rather than serving as substitute proofs for the present manuscript. The primary proofs of the explicit results stated here are in this article. In later editions, stable source identifiers should be used in place of page numbers alone.

A reader who wants to understand…	Where to start here	In the companion volume
Why counting maternal generations is a renewal process	Sections 2–3	M0, p. 10
Why two branches can merge, and what a first collision means	Section 4, Figure 1	M3c, p. 233
What geography and demographic scenarios add	Sections 5, 11	M3, p. 213
How shared ancestors influence a cohort	Sections 8, 9, Figure 2	M4, pp. 481, 485
How mean gaps and range connect to structural marks	Section 10	M5e, p. 1003
How historical inputs could enter a model	Section 11	M6, p. 1035

Companion PDF: Open the 1,406-page full technical volume (from the extracted archive). Some PDF readers restrict links to local files; in that case open the PDF directly from the `TECHNICAL` folder.

The files `proof_navigation/PROOF_MAP.csv` and the technical volume's internal index preserve more detailed citations. The compilation of historical parameters in M6c is a *contract*, not evidence

that human historical pedigrees or the general dense–sparse estimates have been validated. The supplement retains the unresolved arguments transparently, without inserting audit-management labels into this article's mathematics.

A Technical details of the finite-state model

A.1 Proof that ghost labels yield independent collision trials

Every actual vertex has independently assigned maternal age and mother identity; one ancestor's out-edge is never revisited before a coalescence because ages are strictly positive. Up to the first shared identity, the two explored sets of vertices are disjoint. Hence conditional on their age increments the two label sequences are fresh, independent uniform selections. A shared label can arise only when both paths arrive at the same calendar birth date. Conversely, if the labels coincide at that date, both paths have reached the same person and therefore merge permanently in the actual pedigree. The ghost paths can still be continued independently after that stopping date as an *auxiliary* construction; they are not claimed to be jointly consistent with the real pedigree after merger.

A.2 Uniform occupation to conditional Laplace convergence

For a continuous bounded function g supported on $[0, S]$, let $h(u) = 1-e^{-g(u)}$. A partition into finitely many subintervals and monotonicity of M_n imply, from Lemma 3.1,

$$\frac{1}{n}\sum_{1\le t\le nS} I_t h(t/n) \xrightarrow{\mathbb{P}} r^2\int_0^S h(u)\,du. \tag{57}$$

Indeed the integrals of piecewise-constant approximants converge, and the error is at most $\|h - h_{\text{step}}\|_\infty M_n(S)/n$, which is tight. The logarithmic remainder in (15) is bounded by $C\sum I_t/N_n^2 \le CnS/N_n^2 \to 0$. Combining these two observations gives (16) with no missing exchange of a limit and random conditioning.

A.3 The independent limit requires a joint calculation

Consider an exponential time T with rate one, and set $Z = \Phi^{-1}(1 - e^{-T})$. Then Z is standard normal by the probability integral transform, while Z is a deterministic function of T. Thus the separate limits $Z \sim N(0, 1)$ and $T \sim \text{Exp}(1)$ could never imply independence. In Theorem 4.1, the conditional Laplace factorization does exactly the extra work required.

A.4 Why the first-point boundary is harmless here

For any $\eta > 0$, from (17) and the fact that there are at most $n\eta$ potential date comparisons,

$$\mathbb{P}(\tau_n \le \eta) \le \frac{\lfloor n\eta\rfloor}{N_n} \longrightarrow \frac{\eta}{\nu}. \tag{58}$$

Letting $\eta \downarrow 0$ excludes an atom at zero, independent of the vague-convergence argument. The convergence of survival at each fixed $u > 0$ identifies the first-time law. This specifically rules out the spurious conclusion obtainable from only vague convergence on $(0, S)$.

B Finite-state and observation checks

B.1 Residual-chain dynamics and variance check

For the two-state chain of (5), $\pi_0 = 1/(2-p)$ and $\pi_1 = (1-p)/(2-p)$. The centered renewal reward is $f(0) = 1 - r$ and $f(1) = -r$. The Poisson equation $h - Ph = f$ has $h(0) = r$ and $h(1) = 0$. The martingale difference $h(X_t) - (Ph)(X_{t-1})$ is zero in conditional variance if $X_{t-1} = 1$ and has conditional variance $p(1-p)r^2$ if $X_{t-1} = 0$. Thus the asymptotic bracket equals $rp(1-p)r^2 = p(1-p)/\mu^3$. The product-chain renewal coincidence rate is $\pi_0^2 = r^2$. These two constants enter different places: the first is the diffusion coefficient, the second the collision intensity. Confusing them gives an incorrect stopped process.

B.2 A warning about vague convergence of collisions

Point measures $\delta_{1/n}$ on $(0, S)$ converge vaguely to the zero measure on compact subsets bounded away from zero, but their first points tend to zero. Hence convergence of point measures alone on $(0, S)$ does not justify convergence of the *first collision* with cemetery convention. The survival product (17) and the bound in Section A exclude such a boundary mass within the present model. In general spatial environments this needs to be checked again.

B.3 Independent marks versus independent people

In the partition model, independence of block marks is compatible with strong dependence between descendant marks: members of the same ancestral block receive exactly the same value. If the mark law is not independent across blocks, Proposition 8.2 shows the additional covariance terms. The effective number of blocks is not the number of independent sampled leaves. A with-replacement pair samples blocks with probability $H = \sum w_j^2$ of coinciding; without-replacement pairs involve $N/(N-1)$ corrections as in (53).

B.4 A compact map for the companion proofs

The independent technical volume M0–M6c (frozen reader edition, 27 October 2026) contains the full foundational demonstrations. In that specific edition, M0's stationary theorem begins at printed page 10; the unified M3 document at page 213; the final M3c source at page 233; M4 unification at page 481, with a highlighted M4 theorem at page 485; M5e at page 1003; and M6 unification at page 1035. The supplementary file `proof_navigation/PROOF_MAP.csv` records the exact edition and links each paper claim to those source locations, explicitly distinguishing background sources from independently proved claims in the present article. A page pointer is not a substitute for a proof; a different edition must be re-indexed before use.

C Data, software and claims of scope

The numerical examples use synthetic maternal pedigrees and exact finite-state transition matrices. The source code, reference test outputs, revision history, and checksums are bundled with this manuscript. No individually identifiable historical genealogical records are included and none were used to estimate model parameters. The additional full-cohort code uses independent random pedigree generation, exchangeability checks, exact Hoeffding-tail controls and rare-merger enumeration. The accompanying M0–M6c volume preserves the general framework, but an included argument from that volume is not a proof of a claim in this paper unless its assumptions and conclusion are explicitly matched.

The manuscript has received an internal adversarial reading, not external independent peer review. The status of its novelty with respect to all specialist literature, and the dense–sparse full-cohort pro-

gramme, remain open. The author will complete the formal authorship, affiliation and journal-specific declarations before any submission.

D Reproducibility contract

The included standard-library Python scripts check: the residual-chain invariant law and asymptotic variance; the exact finite-n matrix identity against brute-force state-path enumeration; convergence of exact first-collision survival probabilities; exhaustive binary block-mark formulas; exact Bernoulli block expectations; and 2,500 randomized cohort-transfer inequalities. These are computation checks, not replacements for the theorems proved above. The scripts also calculate exact survival probabilities for two finite-regime Cox scenarios, and verify the real genealogical stopping identity on 960 simulated pedigrees. They emit JSON and CSV reports; no restricted historical or human-identifiable data enter any result in this paper.

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